Sensitivity

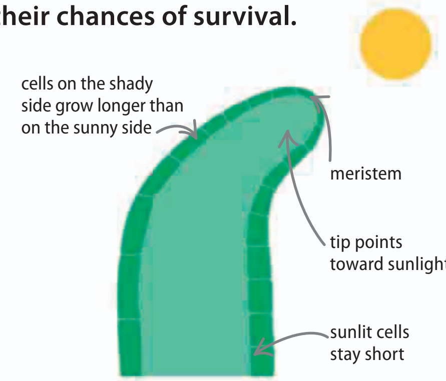
LIVING ORGANISMS SENSE THEIR ENVIRONMENT IN DIFFERENT WAYS.

SEE ALSO	
Human senses	64–65 ❯
Functional groups	160–161 ❯
Electromagnetic waves	194–195 ❯
Light	196–197 ❯
Sound	200–201 ❯

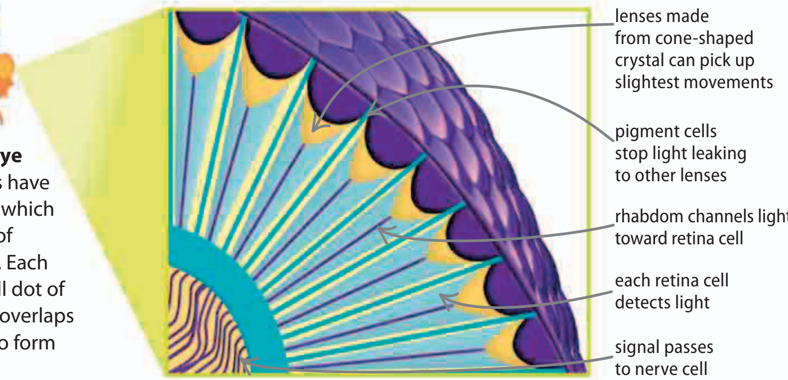
All living things are sensitive to their surroundings, such as changes in light, sound, or chemistry. This sensitivity allows organisms to respond, for example to a threat, increasing their chances of survival.

Tropism

Plants can sense the factors in the environment that help them maximize their growth. This is called tropism. A seed is sensitive to gravity (gravitropism), so its roots grow down into the soil. The roots also turn toward water in the soil (hydrotropism), while the stem grows toward sunlight (phototropism). Phototropism causes a growing point (the meristem) to face the Sun by growing cells on one side of the stem longer than those of the other.



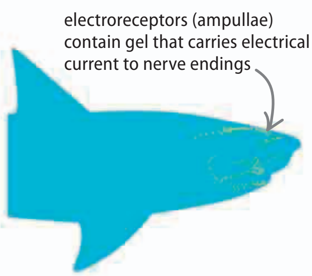
◁ **Phototropism**
Sunlight inhibits the production of growth hormones or auxins. The cells on the shady side of the stem release auxins. That makes the cells in shade grow longer, while the cells on the sunny side stay short.



▷ **Compound eye**
Many arthropods have compound eyes, which have thousands of individual lenses. Each lens forms a small dot of an image, which overlaps with other dots to form a larger image.

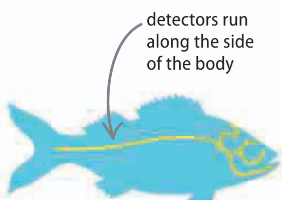
Animal senses

The human senses of touch, smell, sight, hearing, and taste are all used by animals, but not in the same way. For example, a grasshopper hears with pressure-sensitive knees, a housefly tastes its food by standing on it (its taste buds are on its feet), while a moth detects smells with its feathery antennae. Some animals have senses that do not compare to human ones.

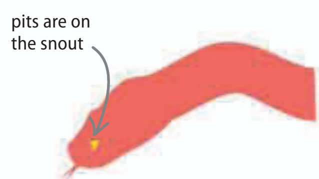


△ **Ampullae of Lorenzini**
Sharks have electroreceptors that pick up the electric fields produced by the muscles of other animals. This allows them to find prey in the dark water.

△ **Whiskers**
Whiskers are ultra-sensitive hairs used by mammals to feel their way in the dark. They are wider than the head, so the animal knows if it is heading into a tight spot.



△ **Lateral line**
Fish use a motion sensor, called the lateral line, running along the side of the body. It picks up the swirling water currents created by other animals moving nearby.



△ **Heat pits**
Pythons and vipers have hollow pits on their snouts that detect the body heat of warm-blooded prey. The pits also warn the snake if it should avoid the other creature.